

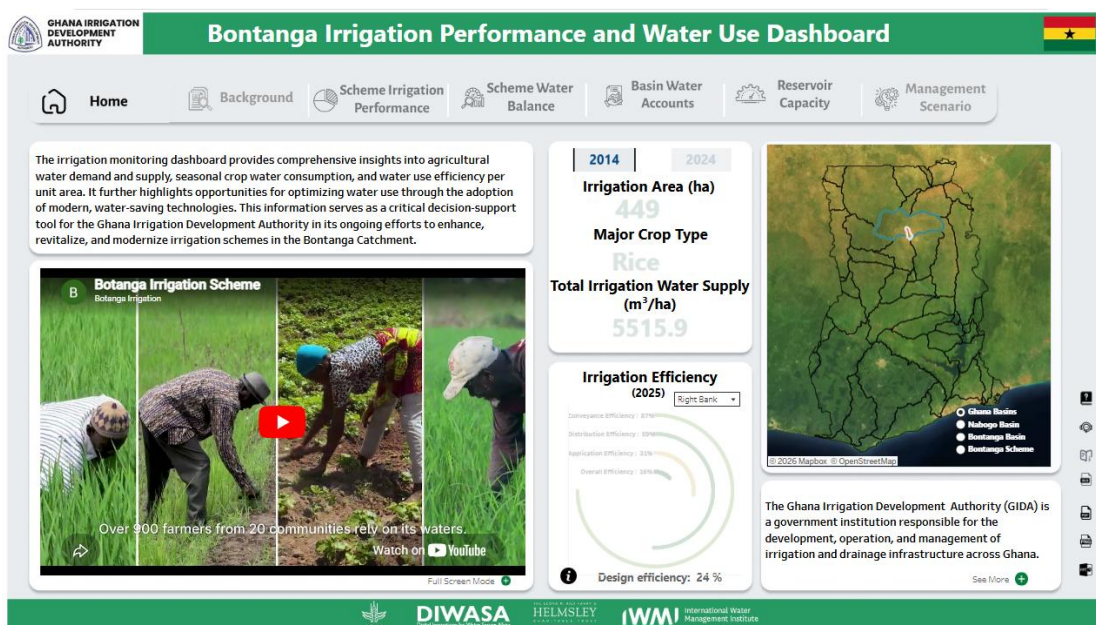
User Manual – Bontanga Irrigation Monitoring Dashboard

01. What is a Irrigation Monitoring Dashboard ?

An **irrigation monitoring dashboard** is a digital tool that displays real-time or historical information about irrigation systems, water usage, and related environmental factors. It helps farmers, water managers, and decision-makers **visualize, track, and manage irrigation efficiently**. The dashboard integrates data from multiple sources, such as weather stations, soil moisture sensors, satellite imagery, and flow meters, and presents it in an easy-to-understand format using charts, maps, and tables.

Key purposes of an irrigation monitoring dashboard:

1. **Monitor Water Usage:** Shows how much water is being applied to fields and helps prevent over- or under-irrigation.
2. **Track Soil and Crop Health:** Uses sensor or satellite data to monitor soil moisture, evapotranspiration, and crop conditions.
3. **Improve Efficiency:** Helps optimize irrigation schedules, saving water and energy while maintaining crop productivity.
4. **Support Decision-Making:** Provides actionable insights for farmers and water managers to plan irrigation, detect leaks, or forecast water demand.
5. **Visualization and Reporting:** Converts complex data into visual charts, graphs, and maps for easy interpretation.



02. Key Elements of Dashboard

The Bontanga Irrigation Monitoring Dashboard is an interactive tool developed to monitor, analyze, and visualize irrigation scheme performance in the Bontanga area. It integrates field observations, hydrological data, and spatial maps to support decision-making for irrigation management.

There are five key elements in the dashboard:

- i) Home
- ii) Background
- iii) Scheme Irrigation Performance
- iv) Scheme Water Balance
- v) Basin Water Accounts
- vi) Reservoir Capacity
- vii) Management Scenario



Figure 1 Dashboard Tabs

01. Home Page – Bontanga Irrigation Monitoring Dashboard

1.1. Overview of the Irrigation Monitoring Dashboard

This dashboard provides key insights into water demand and supply, seasonal crop water use, and water use efficiency. It helps identify opportunities for water savings through modern irrigation practices and supports the Ministry of Irrigation and Lowlands in managing and modernizing irrigation schemes effectively.

The irrigation monitoring dashboard provides comprehensive insights into agricultural water demand and supply, seasonal crop water consumption, and water use efficiency per unit area. It further highlights opportunities for optimizing water use through the adoption of modern, water-saving technologies. This information serves as a critical decision-support tool for the Ghana Irrigation Development Authority in its ongoing efforts to enhance, revitalize, and modernize irrigation schemes in the Bontanga Catchment.

Figure 2 The description of the basin provided in the Irrigation monitoring dashboard overview page

1.2 Video Section

Embedded video explaining the Bontanga irrigation system, providing users with a visual and verbal overview of irrigation management practices.



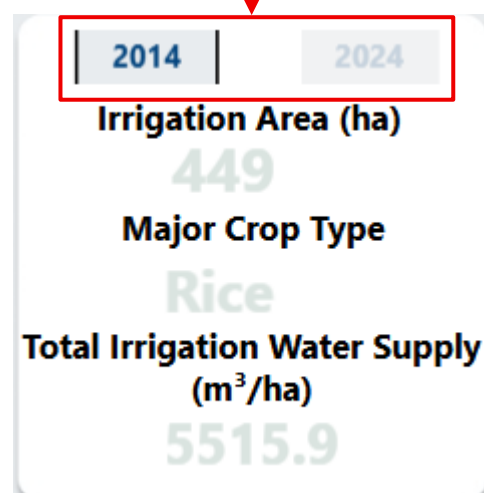
1.3 Key Indicators

The Key Indicators section serves as a high-level summary of the most critical metrics for monitoring and managing irrigation systems. It is designed to provide users with immediate insights into the status of irrigated agriculture, helping to make informed decisions efficiently.

View video in full screen mode

- **Irrigation Area (ha):** Represents the total land area currently receiving irrigation. This metric is crucial for understanding the extent of irrigated farming, planning water allocation, and assessing the coverage of irrigation infrastructure. It also helps in monitoring expansion or reduction of irrigated areas over time.
- **Major Crop Type:** Displays the dominant crop cultivated within the irrigated area. Knowing the main crop type is essential because different crops have varying water requirements, growth cycles, and sensitivities to water stress. This information helps optimize irrigation schedules and prioritize water resources according to crop needs.
- **Total Irrigation Water Supply (m^3/ha):** Indicates the amount of water supplied per hectare of irrigated land. This metric reflects water use efficiency, adequacy of supply, and helps identify areas that may be under- or

Click here change years

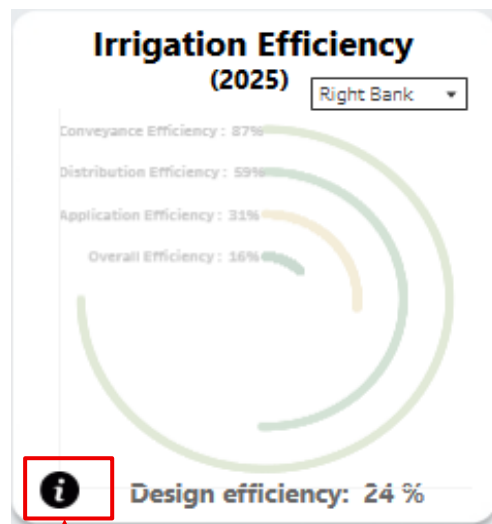


over-irrigated. By tracking this indicator, users can ensure sustainable water management and prevent water wastage.

1.4 Irrigation Efficiency Gauge

Visualizes overall irrigation efficiency and its components, including application, conveyance, and distribution efficiencies, helping users understand how effectively water is being used.

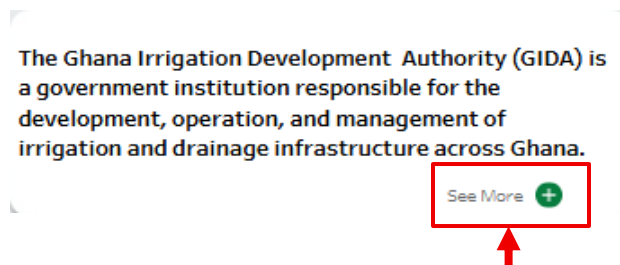
01. **Design efficiency** refers to the efficiency parameters that are targeted and selected by designer when planning and creating an irrigation system.
02. **Conveyance efficiency** reflects how effectively water is delivered from its source to the farm gate, highlighting any losses that occur during transport.
03. **Application efficiency**, by contrast, measures how much of the applied irrigation water is stored in the root zone and available to crop.
04. **Overall efficiency** is the multiplication of Conveyance efficiency, application and distribution efficiency.



Click for definition and more information

1.5 Ministry Information Box

Provides background about the Ethiopian Ministry of Irrigation & Lowlands, its establishment, and its role in sustainable irrigation and improving lives in lowland areas.



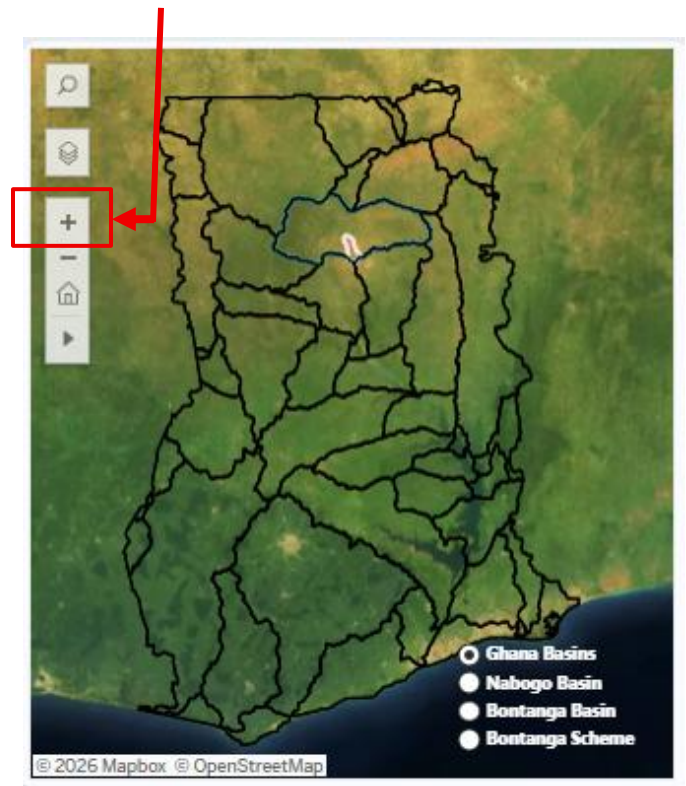
Click here to direct MILLS website

Figure 3 MILLS Information

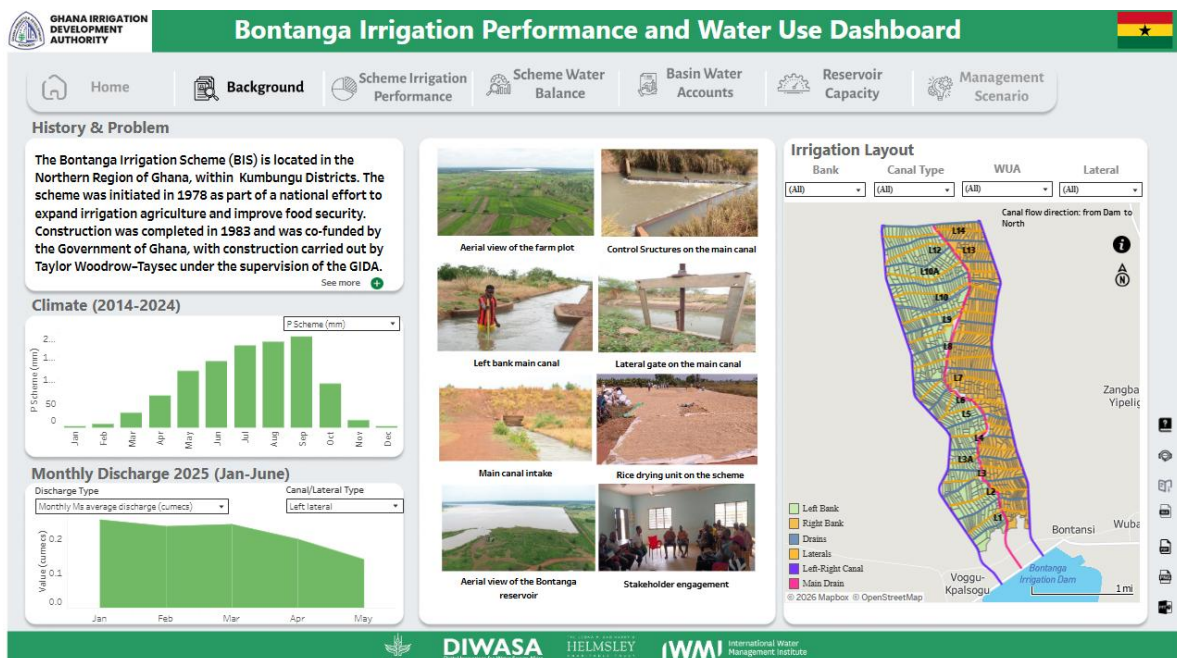
1.6 Map Location

The interactive map displays the location of the Bontanga irrigation scheme within the Awash Basin and Ethiopia. Users can zoom in and out or pan across the map to view specific areas, making it easier to understand the spatial distribution of water resources, irrigation schemes, and surrounding basins.

Click Zoom to Bontanga



02. Background page

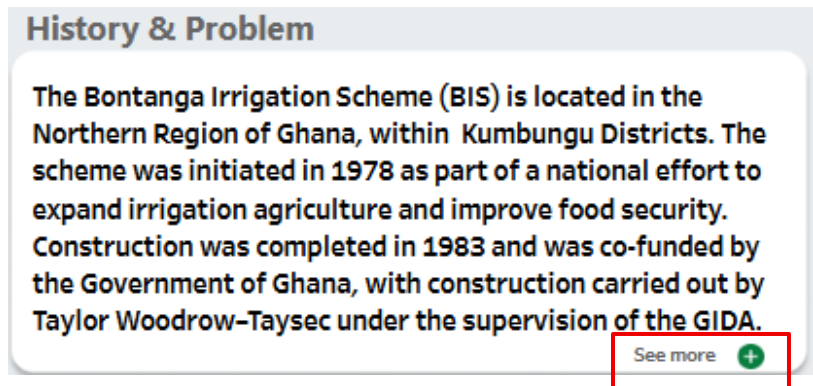


2.1 History and Problem

The **History and Problem** section of the dashboard provides users with essential background information about the

Bontanga Irrigation Scheme. It outlines the origins of the scheme, which was initiated in 1978 and completed in 1984, and describes

how it was co-financed by the Ethiopian government along with the World Bank, African Development Bank, and European Economic Community. The section also notes that construction was carried out by Ethiopian contractors under the supervision of Sir William Halcrow & Partners Ltd. In terms of scale, the scheme covers nearly 15,000 hectares, of which approximately 10,300 hectares are irrigated through gravity while the rest depend on pumping. In addition to presenting this historical context, the section highlights ongoing challenges such as waterlogging, the operational demands of pumping systems, and related management issues. Users can access this section to gain an overall understanding of the scheme's development, scope, and the key problems that continue to affect its performance.



Click Here for Navigate to more history

2.2 Climate Data Bar Chart (2010–2023 Average)

By visualizing long-term average climate data, this chart helps farmers, water managers, and planners anticipate **seasonal water availability, irrigation demand, and crop stress periods** to improve water-use efficiency in the irrigation scheme.

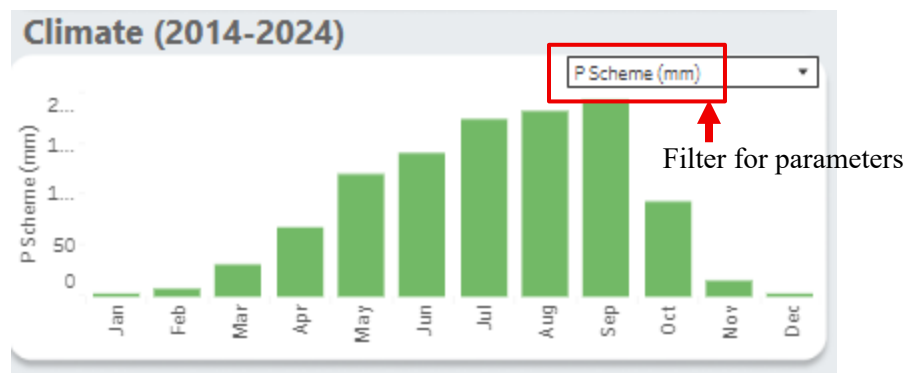
This interactive bar chart presents key climate indicators averaged across the years **2010 to 2023**, displayed on a **monthly basis**. Users can explore seasonal climate patterns relevant to irrigation planning.

- **Rainfall:** Shows the monthly average rainfall, helping to understand wet and dry periods.

- **Maximum Temperature (Tmax):** Displays average monthly peak temperatures, useful for analyzing heat stress on crops.
- **Minimum Temperature (Tmin):** Highlights the average lowest monthly temperatures, providing insights into cooler periods.
- **Reference Evapotranspiration (ET0):** Indicates the water demand of crops under standard conditions, supporting irrigation requirement assessments.

Interactive Features:

- All climate parameters are available in a **single plot**.
- A **filter option** allows users to select and view any parameter individually or compare them dynamically.
- This flexibility supports decision-making by enabling users to focus on specific climate drivers of crop water needs.



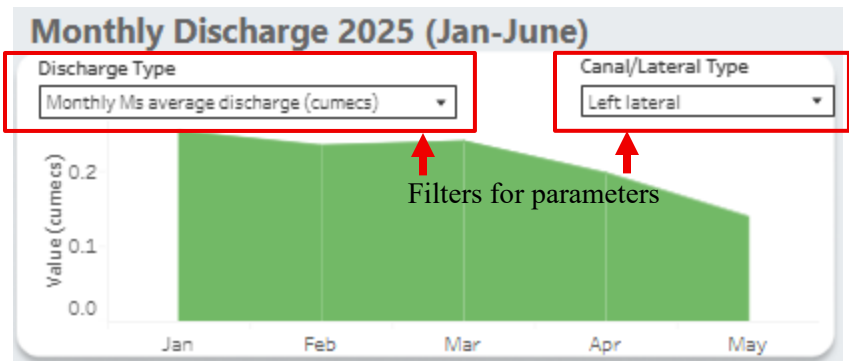
2.3 Discharge at Weir Site (1991–2024)

This chart displays long-term discharge data at the weir site, covering the period **1991 to 2024**. It provides key flow indicators that are critical for understanding water availability for irrigation and system management.

- **80% Dependable Flow:** Represents the minimum flow that can be expected 80% of the time, ensuring reliable planning for irrigation under varying hydrological conditions.
- **Weir Intake Capacity:** Shows the designed intake capacity of the weir structure, indicating how much water can be diverted for irrigation purposes.
- **Awash Flow at Weir Site:** Displays the actual river flow measured at the weir location, helping to compare natural flow against intake capacity and dependable flow levels.

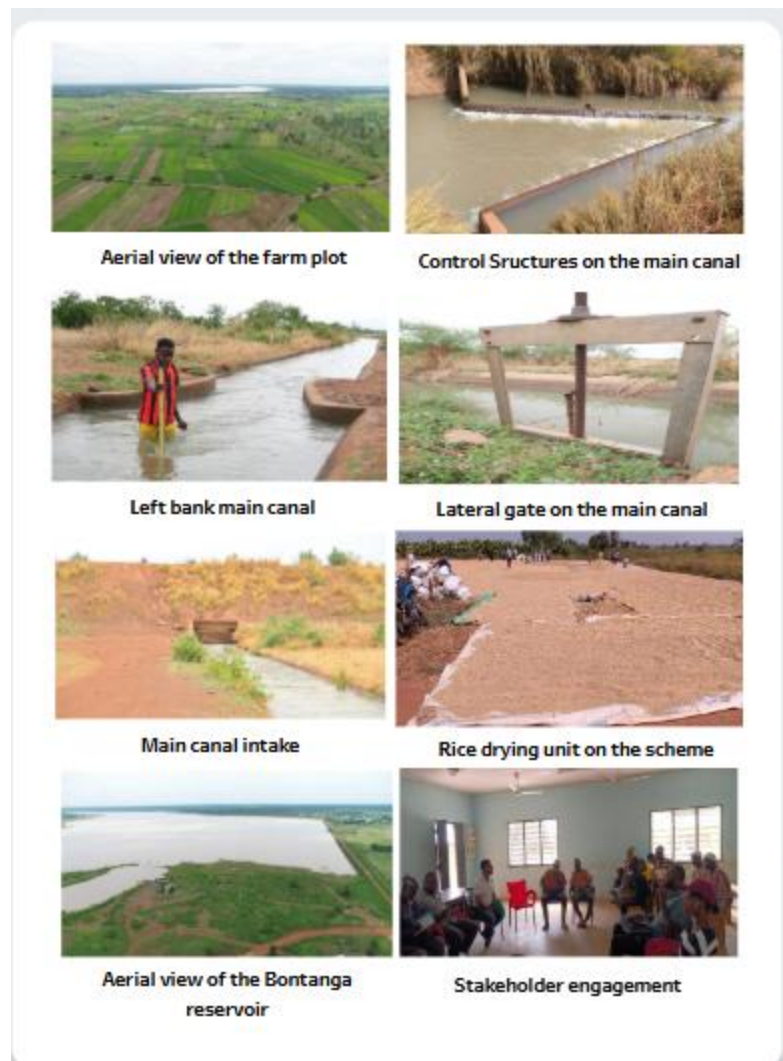
Interactive Features:

- Users can apply **filters** to switch between different flow parameters.
- The chart enables direct **visual comparison** between dependable flow, intake capacity, and actual river discharge over time.



2.4 Photo Gallery

This section provides a visual overview of the irrigation scheme and related field activities. Users can **click on any image** to view it in **full size**, allowing for a closer look at infrastructure, field conditions, and ongoing operations.



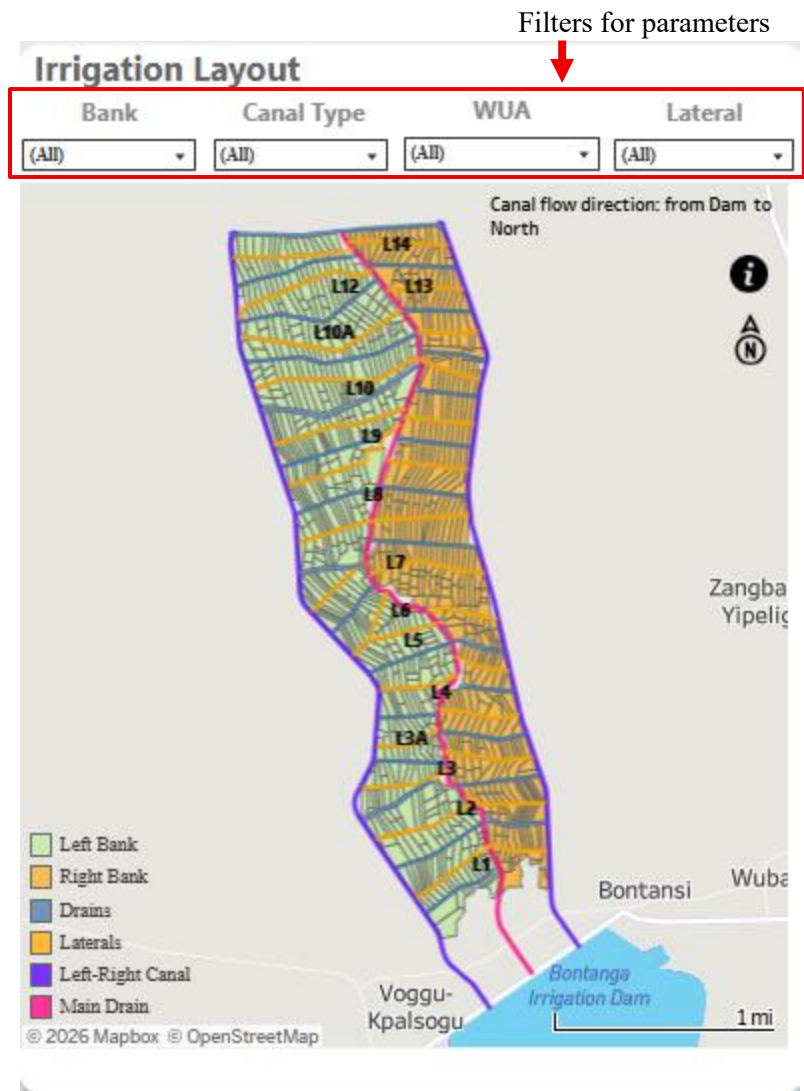
2.5 Irrigation Layout

The **Irrigation Layout** map provides a spatial overview of the irrigation scheme, showing the distribution of **blocks, canal types, and water sources**.

- Users can **filter** by these three parameters to view specific components of the irrigation system.
- By clicking on a block, users can access a **popup with detailed information** about that block (e.g., area, crops, water allocation).

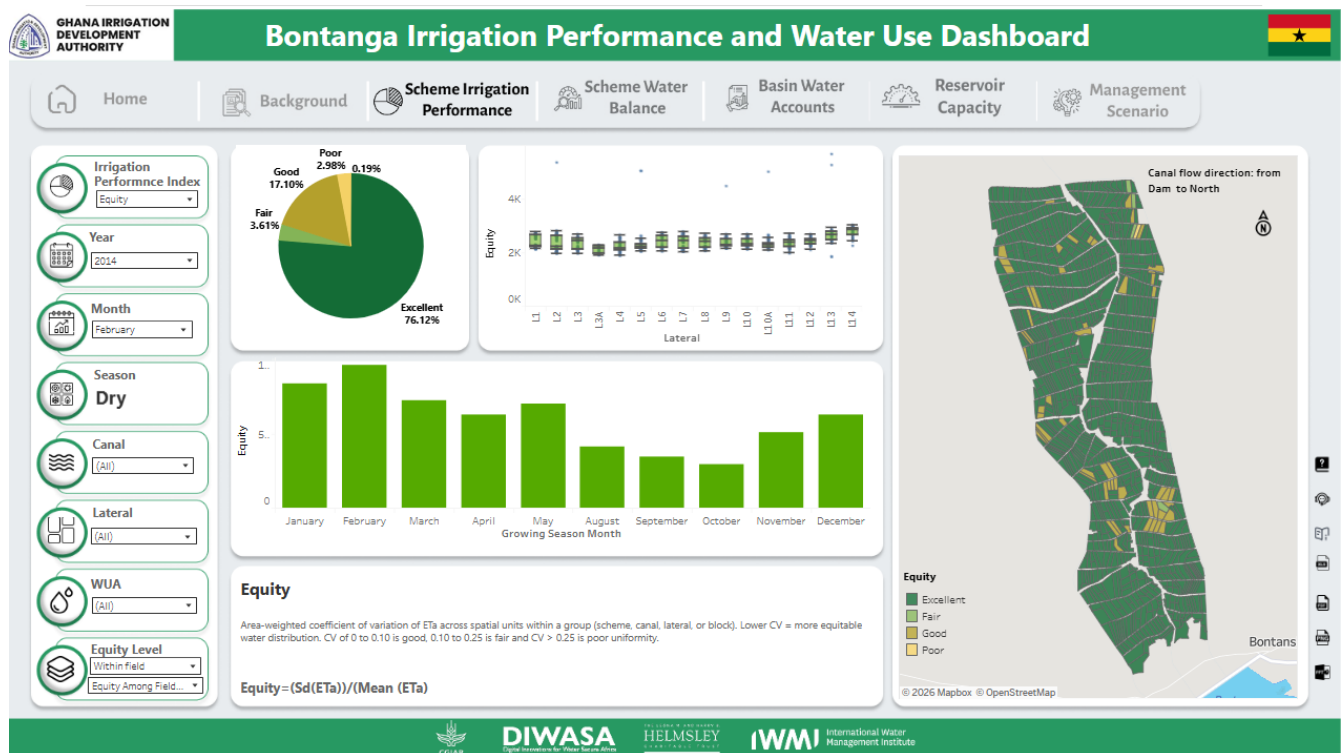
This interactive map helps managers and stakeholders understand how water is distributed and managed within the scheme.

- ✓ **Irrigation Layout:** A schematic or map that shows the arrangement of irrigation infrastructure, including canals, blocks, and water sources, designed to manage and distribute water efficiently.
- ✓ **Gravity Irrigation System:** A method of conveying water to fields using natural ground slope, without the need for pumping, making it energy-efficient and cost-effective.
- ✓ **Pumping Station:** An engineered facility equipped with pumps that lifts water from rivers, reservoirs, or underground sources to supply the irrigation system where gravity flow is not sufficient.
- ✓ **Main Canal:** The primary channel that conveys water from the water source (weir, reservoir, or pumping station) into the irrigation network.



- ✓ **Secondary Canal:** Smaller channels branching from the main canal to deliver water to specific irrigation blocks or fields.
- ✓ **Drain:** A channel designed to remove excess or waste water from the irrigation area, preventing waterlogging and salinity.
- ✓ **Secondary Drain:** Smaller drainage channels that connect local field drains to the main drain for effective removal of surplus water.

03. Scheme Irrigation Performance



3.1 Filters & Controls

The left panel of the dashboard provides several filters that allow users to customize the displayed data according to their needs. The **Irrigation Performance Index** can be analyzed through different parameters including **Equity**, **Relative Evapotranspiration**, **Relative Irrigation Supply**, and **Irrigation**. These indicators help assess the efficiency and fairness of water distribution across the scheme. Users can also select the **Year** of analysis, with available data ranging from **2010 to 2024**, making it possible to compare historical and recent irrigation performance. The **Crop Type** filter allows users to focus on specific crops such as **cotton** and **vegetables**, which are the main cultivated crops in the Bontanga scheme. Additionally, the **Canals**

filter enables users to either analyze all canals together or select a specific canal for detailed monitoring. Finally, the **Season** filter provides the option to view performance during the Primary Season (May–October) allowing for seasonal comparisons of irrigation practices and water availability.

Select a parameter,
year and crop and
canal

3.2 Definition and Fraction of Each Parameter

This section provides a brief explanation and fractional representation of each parameter used in the analysis. The

definition describes what the parameter measures or represents, while the **fraction** indicates its proportional contribution or weight in the dataset. Together, these help in understanding the role and significance of each parameter in evaluating variations during the selected crop-growing season.

Equity

Area-weighted coefficient of variation of ETa across spatial units within a group (scheme, canal, lateral, or block). Lower CV = more equitable water distribution. CV of 0 to 0.10 is good, 0.10 to 0.25 is fair and CV > 0.25 is poor uniformity.

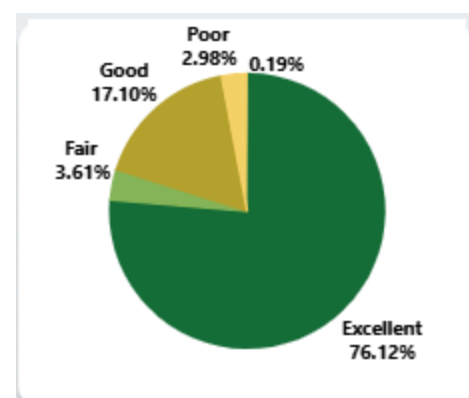
$$\text{Equity} = (\text{Sd}(\text{ETa})) / (\text{Mean}(\text{ETa}))$$

The **definition section** located below the visualizations provides a clear explanation of the selected **Irrigation Performance indicators**, helping users interpret what the displayed values represent

3.3 Pie chart

This pie chart illustrates the percentage distribution of each parameter across different levels. It visually represents how each parameter contributes proportionally within the dataset, making it easier to compare their relative significance and identify which levels dominate or have lesser representation.

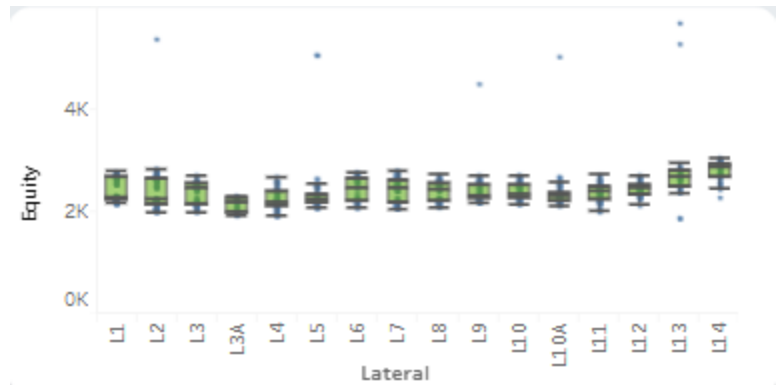
This chart is fully **interactive and dynamically linked to the filters** on the left panel. When users adjust filters such as **Indicator, Year, Crop Type,**



Canal, or **Season**, the chart automatically updates to display values relevant to the selected conditions.

3.4 Box plot

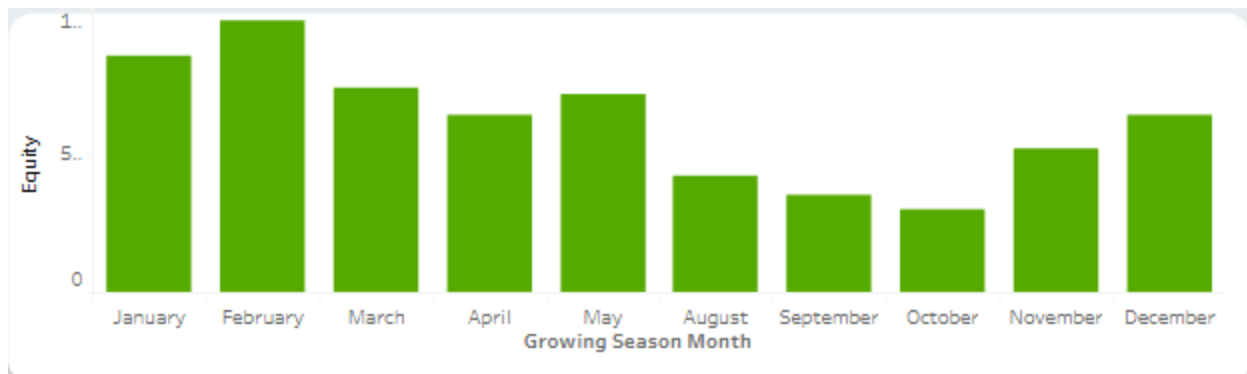
This box plot displays the distribution of each parameter across different canals. It shows how parameter values vary canal-wise, highlighting the median, quartiles, and possible outliers. This visualization helps compare variations and identify trends or anomalies among the canals.



This chart is fully **interactive and dynamically linked to the filters** on the left panel. When users adjust filters such as **Indicator**, **Year**, **Crop Type**, **Canal**, or **Season**, the chart automatically updates to display values relevant to the selected conditions.

3.5 Bar Chart

This bar chart illustrates the variation of each parameter during the selected crop-growing season. It highlights how parameter values change over time within the season, allowing comparison between different parameters and identifying key trends that influence crop growth.



This chart dynamically updates based on the selected **Indicator**, **Year**, **Canal**, and **Crop Type** filters.

3.6 Map

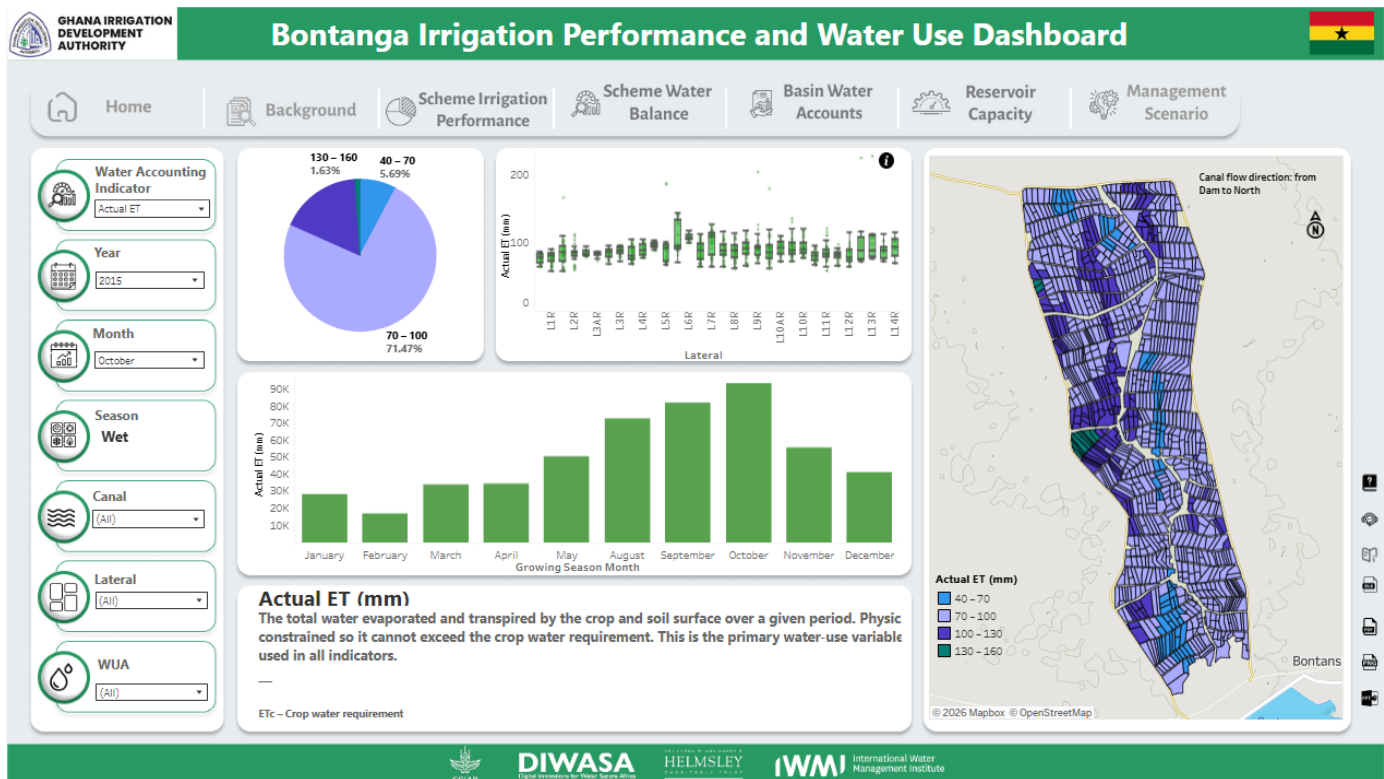
The map section is designed to display the spatial distribution of all parameters with interactive filtering options. Users can dynamically explore the data by selecting specific **parameters**, **years**, and **crop types**. As filters are applied, the map automatically updates to reflect the relevant data, allowing users to visualize how each parameter changes geographically across different conditions and time periods.

Map Section **Indicator**, **Year**, **Canal**, and **Crop Type** are fully interactive and directly control what is displayed on the **map**.



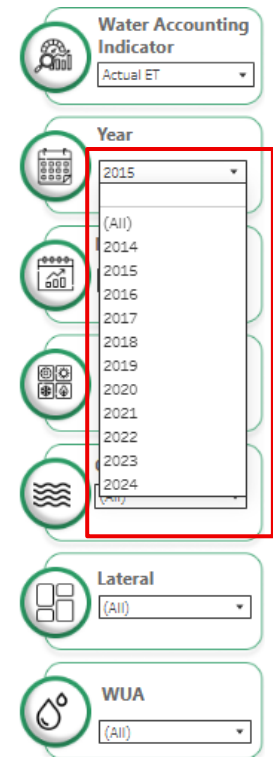
4. Scheme Water Balance

This dashboard provides an interactive visualization of Water Accounting indexes across the Bontanga command area. It allows users to monitor crop water use, canal performance, and spatial distribution of water accounting indicators such as **Green ET** ,**Blue ET** and **Beneficial fraction**.



4.1 Filters & Controls

The left panel of the dashboard provides several filters that allow users to customize the displayed data according to their needs. The **Water Accounting indexes** can be analyzed through different parameters including **Blue Et, Green ET, And Beneficial fraction** . These indicators help assess the efficiency and fairness of water distribution across the scheme. Users can also select the **Year** of analysis, with available data ranging from **2010 to 2024**, making it possible to compare historical and recent irrigation performance. The **Crop Type** filter allows users to focus on specific crops such as **cotton** and **vegetables**, which are the main cultivated crops in the Bontanga scheme. Additionally, the **Canals** filter enables users to either analyze all canals together or select a specific canal for detailed monitoring. Finally, the **Season** filter provides the option to view performance during the Primary Season (May–October) allowing for seasonal comparisons of irrigation practices and water availability.



Water Accounting Indicator
Actual ET

Year
2015
(All)
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
Year

Lateral
(All)

WUA
(All)

4.1 Definition and Fraction of Each Parameter

This section provides a brief explanation and fractional representation of each parameter used in the analysis. The **definition** describes what the parameter measures or represents, while the **fraction** indicates its proportional contribution or weight in the dataset. Together, these help in understanding the role and significance of each parameter in evaluating variations during the selected crop-growing season.

The **definition section** located below the visualizations provides a clear explanation of the selected **Water Accounting Indicator**, helping users interpret what the displayed values represent.

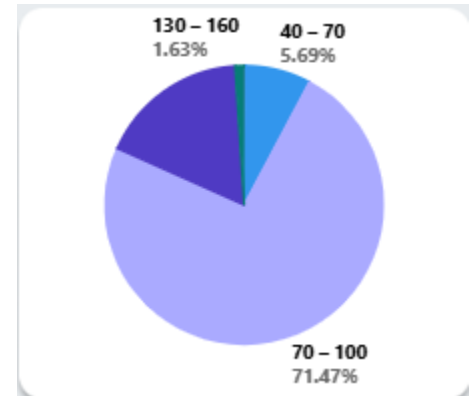
Actual ET (mm)

The total water evaporated and transpired by the crop and soil surface over a given period. Physic constrained so it cannot exceed the crop water requirement. This is the primary water-use variable used in all indicators.

—
ETc – Crop water requirement

4.2 Pie Chart

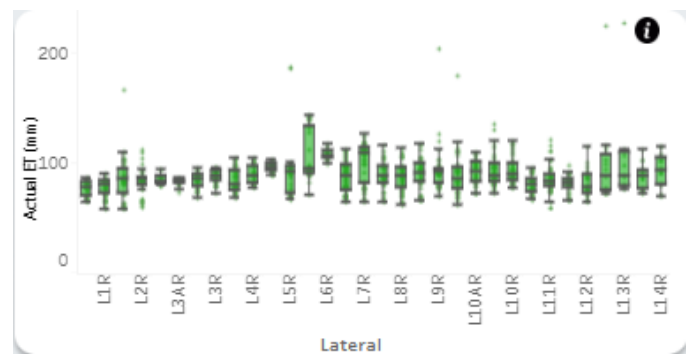
This pie chart illustrates the proportion of different crops cultivated each year, showing their respective areas in hectares. It provides a visual comparison of how crop cultivation varies annually, highlighting which crops occupy larger or smaller areas within the selected years.



4.3 Box plots

This box plot displays the distribution of each parameter across different canals. It shows how parameter values vary canal-wise, highlighting the median, quartiles, and possible outliers.

This visualization helps compare variations and identify trends or anomalies among the canals.



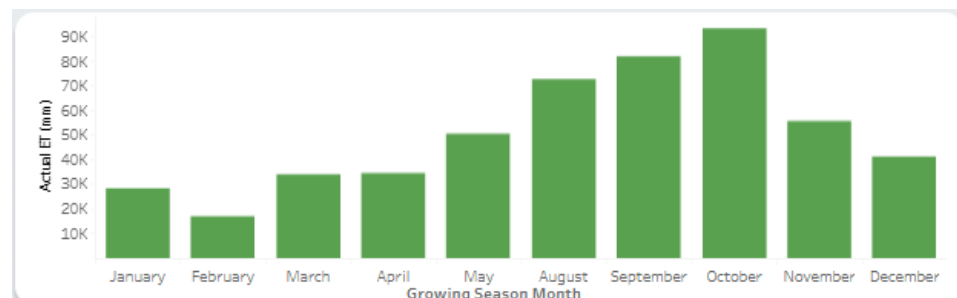
This chart is fully **interactive and dynamically linked to the filters** on the left panel. When users adjust filters such as **Year**, **Crop Type**, **Canal**, or **Season**, the chart automatically updates to display values relevant to the selected conditions. For instance, filtering by a specific crop like *Cotton* or *Vegetables* will show how that particular crop's water use pattern changes, while selecting a specific canal will highlight spatial variations in evapotranspiration within that area. This interactive behavior helps users analyze seasonal water consumption trends.

4.5 Bar Chart

This bar chart illustrates the variation of each parameter during the selected crop-growing season. It highlights how parameter values change over time within the season, allowing comparison between different parameters and identifying key trends that influence crop growth.

This chart dynamically updates based on the selected **Water Accounting Indicator**, **Year**, **Canal**, and

Crop Type filters. When a user changes the **indicator**, the chart reflects the

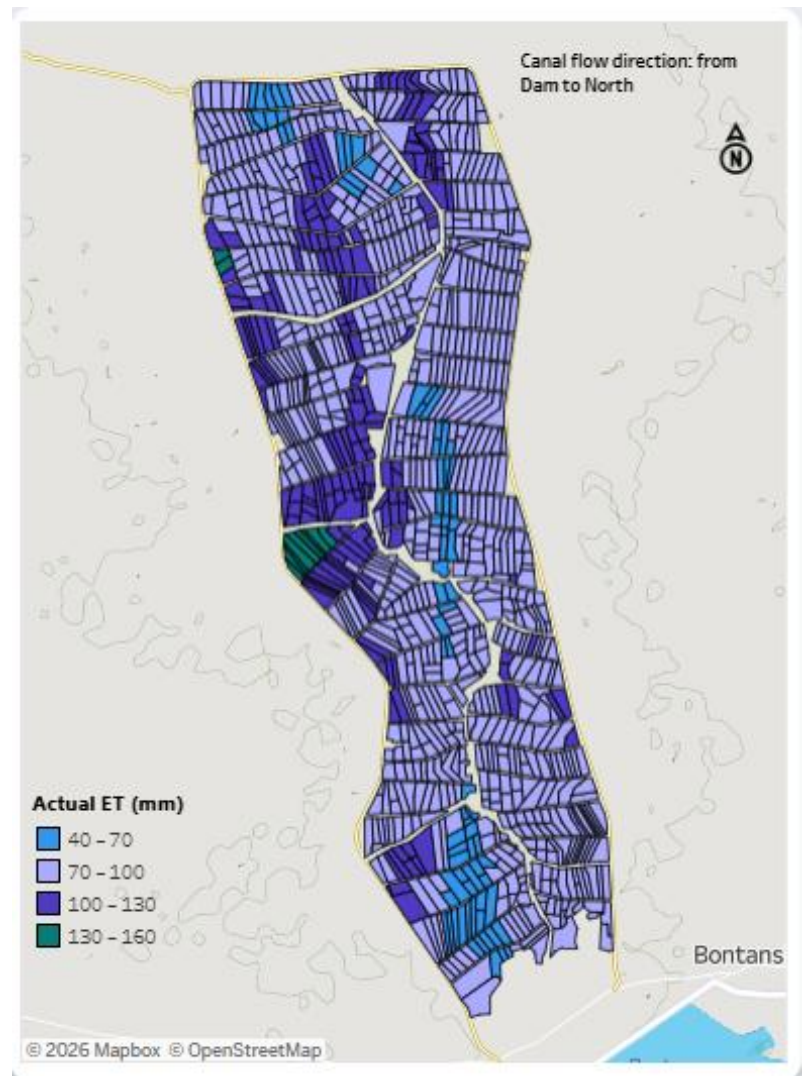


corresponding type (e.g., Green ET). Selecting a specific **year** refreshes the data for that monitoring period. Similarly, choosing a particular **crop type** (e.g., Cotton or Vegetables) tailors the chart to show the pattern for that crop alone. This interactive linkage between the chart and filters allows users to effectively analyze.

4.6 Map

The map section is designed to display the spatial distribution of all parameters with interactive filtering options. Users can dynamically explore the data by selecting specific **parameters, years, and crop types**. As filters are applied, the map automatically updates to reflect the relevant data, allowing users to visualize how each parameter changes geographically across different conditions and time periods.

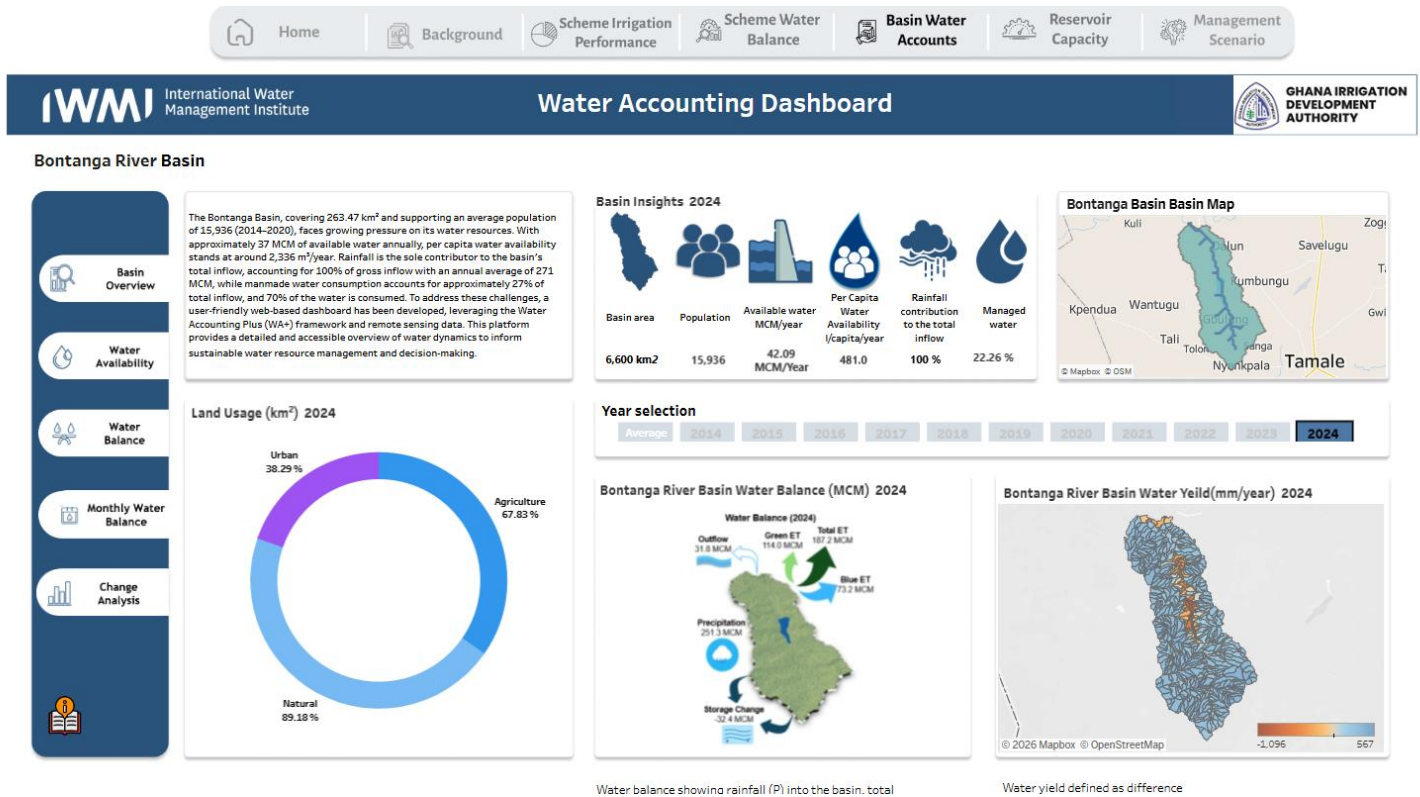
The **filters** on the left panel of the dashboard namely **Water Accounting Indicator, Year, Canal, and Crop Type** are fully interactive and directly control what is displayed on the **map**. When a user selects a specific **indicator** the map updates to visualize that parameter spatially across the irrigation command area using a color gradient to represent different value ranges. Choosing a particular **year** refreshes the map to show data corresponding to that time period. Similarly, the **Canal** filter narrows down the map to display only the selected canal's command area. The **Crop Type** filter lets users focus on specific crops (such as Cotton or Vegetables).



6. Basin Water Accounts

The **Basin Water Accounts** tab serves as a comprehensive, centralized dashboard providing a complete and detailed overview of the selected river basin's water assets, usage, and spatial characteristics. By navigating to this page, users can interact with an all-inclusive analytical layout designed to simplify complex hydrological data into actionable insights for sustainable water resource management.

This page acts as a "dashboard within a dashboard," integrating spatial maps, key performance indicators (KPIs), temporal filters, and land-use metrics in one unified view.



7. Reservoir Capacity

The surface area of the Bontanga Reservoir was estimated using multi-temporal satellite imagery from Sentinel-2 and Sentinel-1 for the period 2018–2024. All preprocessing and initial analyses were conducted within the Digital Earth Africa (DE Africa) cloud-computing environment using Google Colab, while spatial corrections and final vector processing were performed in QGIS.

Sentinel-2 multispectral imagery was used to delineate reservoir water extent under cloud-free conditions. The Normalized Difference Water Index (NDWI) was computed from the green and near-infrared bands to enhance open water features. An empirically defined threshold was applied to the NDWI images to generate binary water masks representing the reservoir surface. Monthly mean water surface areas were derived for each year from 2018 to 2024 to capture temporal variability in reservoir extent. The extracted water masks were exported to QGIS for further refinement. Scenes affected by cloud contamination, atmospheric distortion, or anomalous classification were visually inspected and either corrected or excluded from the analysis. Polygon boundaries were cleaned to ensure realistic reservoir outlines, after which surface areas were calculated for each monthly observation.

Sentinel-1 synthetic aperture radar (SAR) data were also used to extract reservoir water surface area. Speckle noise was reduced using a Lee filter, and the cross-polarized VH backscatter values were converted from an incidence angle to decibel (dB) units. Water bodies were identified by applying a threshold to the VH backscatter. Derived water masks were similarly imported into QGIS for manual refinement and validation of reservoir boundaries.

Final reservoir surface area estimates from both Sentinel-2 and Sentinel-1 datasets were used to estimate storage capacity variations over time separately.

Reservoir storage capacity was estimated using a quadratic area–volume relationship developed from the area–elevation data reported by Mwanga et al. (2024) on the Bontanga reservoir. The relationship between area–elevation was defined using a quadratic linear equation as:

$$V = 4 \times 10^{-7} A^2 + 0.5292A + 13623$$

Where V represents reservoir storage volume (m^3) and A is the water surface area (m^2). This equation was applied to the monthly mean surface areas to estimate reservoir volume and generate the storage capacity curve for Bontanga Reservoir. The **Reservoir Capacity** tab provides an interactive evaluation of the Bontanga Reservoir’s physical dimensions, water storage levels, and historical boundaries using satellite-derived data.

1. Interactive Time-Series Filters (Year & Month Selection)

Located in the upper-middle card, these dropdown menus are the primary controls for the page:

- **Month & Year Dropdowns:** Users can select a specific month and year (e.g., February 2024) to query the historical database.
- **Dynamic Data Updates:** Adjusting these filters instantly recalculates and updates the adjacent metrics, including the season classification (e.g., **Dry** or **Wet**), calculated surface **Area (m^2)**, and estimated **Volume (m^3)** for that specific point in time.

February

February

2024

2024



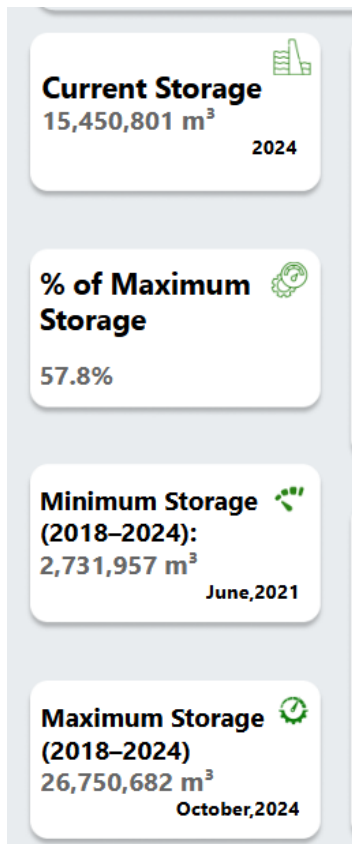
Dry

Area (m^2)

19,194,603

Volume (m^3)

47,069,774



2. Left-Hand Key Performance Indicators (KPI Cards)

These cards track real-time and historical benchmarks for the reservoir:

- **Current Storage (m^3):** Displays the total volumetric water storage calculated for the selected active year.
- **% of Maximum Storage:** A percentage indicator showing how full the reservoir is relative to its absolute maximum capacity.
- **Minimum Storage (2018–2024):** A baseline card highlighting the historical lowest water volume recorded, alongside the exact month and year it occurred (e.g., June 2021).
- **Maximum Storage (2018–2024):** A benchmark card highlighting the peak historical water volume capacity reached, alongside the exact month and year it occurred (e.g., October 2024).

3. Reservoir Storage Capacity Estimation (Methodology Narrative)

- This text pane provides a brief technical summary of how the backend calculations are performed. It notes the use of multi-temporal **Sentinel-1** and **Sentinel-2** satellite imagery processed in **QGIS** and **Digital Earth Africa** using index thresholding (NDWI).
- It highlights the exact quadratic area-volume mathematical relationship used to model the water depth:

$$V = 4 \times 10^{-7} A^2 + 0.5292A + 13623$$

Bontanga Reservoir Storage Capacity Estimation

The surface area of the Bontanga Reservoir was estimated using multi-temporal Sentinel-2 and Sentinel-1 imagery from 2018–2024 within the Digital Earth Africa environment and refined in QGIS. Sentinel-2 imagery was processed using the Normalized Difference Water Index (NDWI), while Sentinel-1 SAR data were processed using VH backscatter thresholding after speckle filtering to extract reservoir water extent. Monthly mean surface areas were calculated and used to estimate storage capacity variations over time.

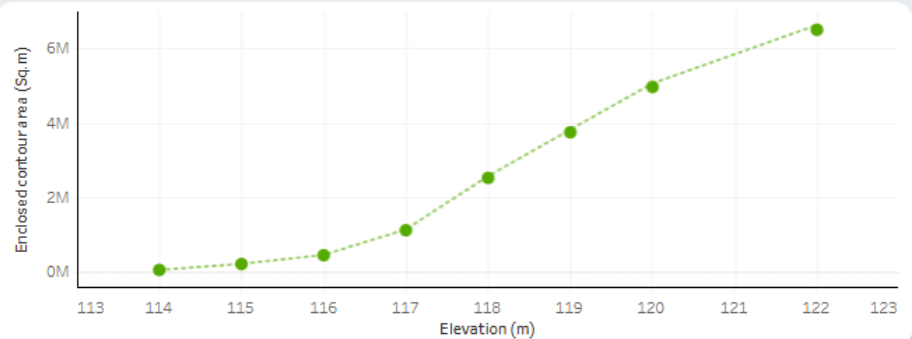
Reservoir volume was estimated using the quadratic area & volume relationship developed by Mwanga et al. (2024):

$$V = 4 \times 10^{-7} A^2 + 0.5292A + 13623$$

4. Enclosed Contour Area Chart (Sq.m vs. Elevation)

- A line graph plotting **Enclosed contour area (Sq.m)** against **Elevation (m)**.
- This curve illustrates the topography of the reservoir bed, mapping out how the surface area expands as the water level rises to higher elevations.

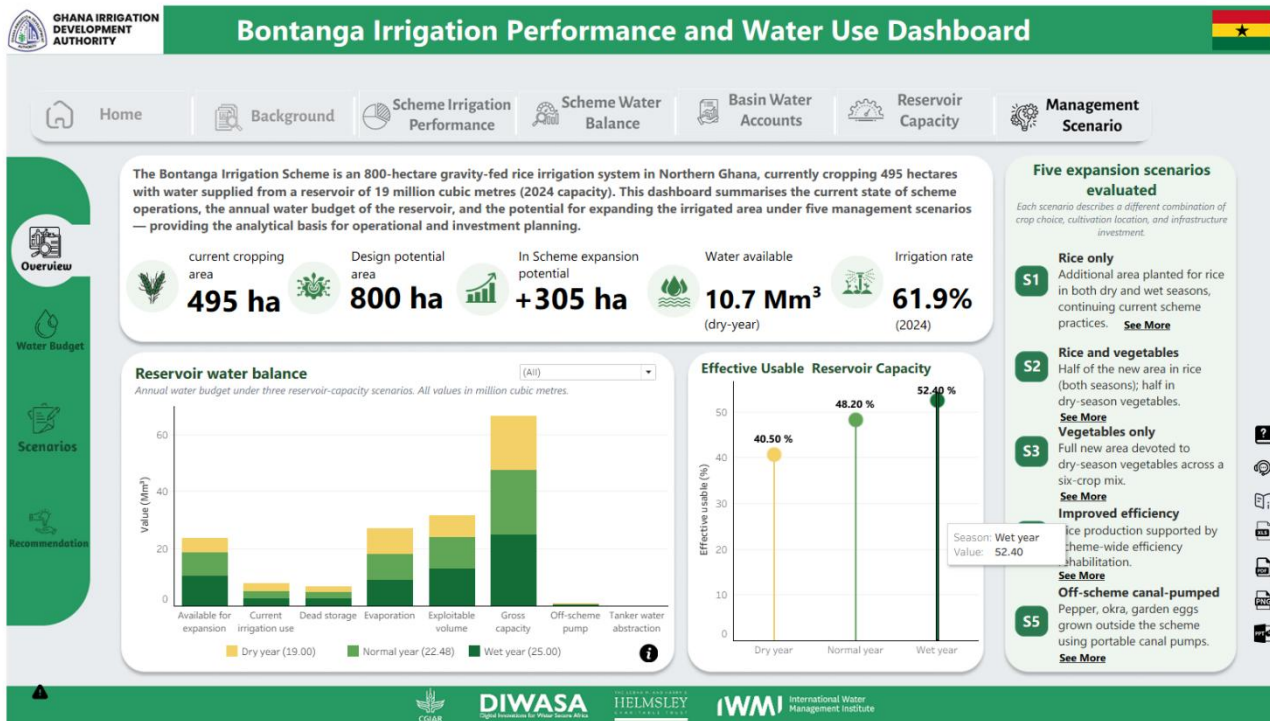
Enclosed contour area (Sq.m)



5. Interactive Reservoir Map Pane

- A high-resolution satellite imagery backdrop overlaying the localized spatial footprint of the reservoir.
- The green shaded vector layer dynamically updates its shape and boundaries to visually represent the actual water extent corresponding to the selected month and year filter.

8. Management Scenario



Management Scenario: Vertical Sub-Navigation Menu

The green vertical sidebar shown in image_16b6e4.png is the dedicated navigation panel for the Management Scenario main tab. It allows users to cycle through different analytical depths of the simulation tool—moving from high-level summaries down to concrete investment suggestions.

Menu Structure and Navigation Modes

Overview (Active Selection): As shown in the image with the white highlighted circular background, this acts as the landing page for the module. It gives users a summary of the scheme's physical dimensions (current vs. design crop areas) and sets the baseline for the expansion potential.

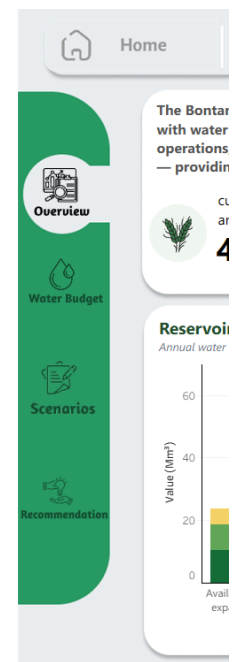
Water Budget: Clicking this item transitions the user to the historic allocation analysis. It reveals how the basin's water volumes behave across ten-year timelines and details natural losses like reservoir evaporation.

Scenarios: This option opens the core simulation matrix where users evaluate the five specific, alternative agriculture expansion strategies (\$S1\$ through \$S5\$), exploring alternative crop combinations like rice variations versus high-value vegetable mixes.

Recommendation: The final option provides a synthesis layer based on the simulated findings. It highlights optimized action paths and efficiency metrics to guide planners toward the most sustainable infrastructure investments.

User Experience Highlight

User Note: This menu functions as a sequential workflow. A user should ideally move from Overview to establish context, check the Water Budget constraints, evaluate alternative expansion options under Scenarios, and ultimately view the data-driven path forward in Recommendation.

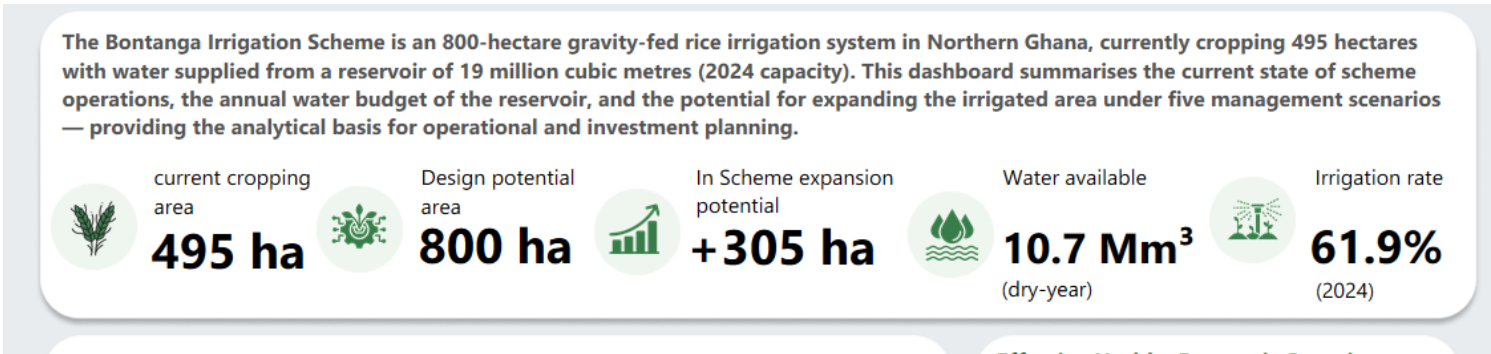


What Can a User See and Interact With?

1. Baseline Performance Metrics (The Top Cards)

At a glance, a user can immediately evaluate the current operational footprint of the irrigation scheme:

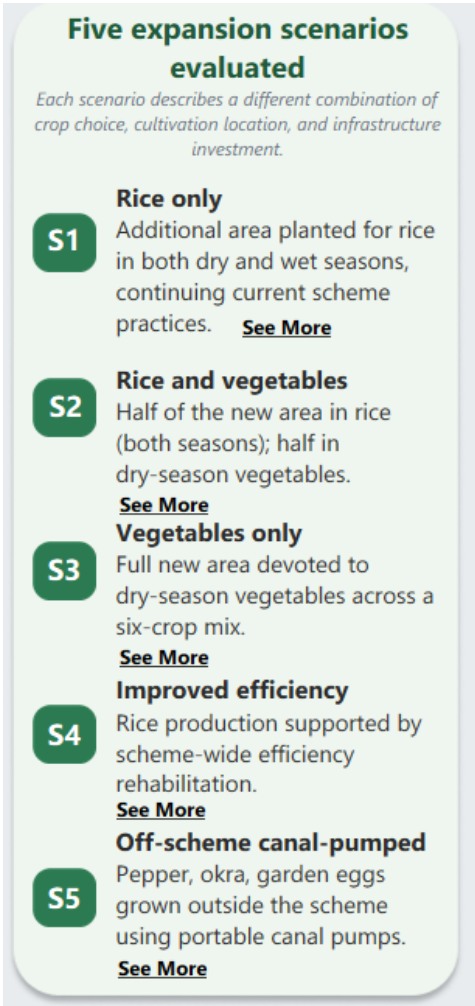
- **Current Cropping Area vs. Design Potential:** Users can see that while the system is designed to handle **800 ha**, it currently only crops **495 ha**.
- **Expansion Potential:** The system automatically highlights an in-scheme expansion capacity of **+305 ha**.
- **Water Allocation Baseline:** Displays critical reference metrics like dry-year water availability and the current irrigation rate (**61.9%**).



Interactive Scenario Evaluator (The Right Sidebar)

This is the core decision-making element. Users can review **five alternative expansion scenarios** labeled **S1 through S5**:

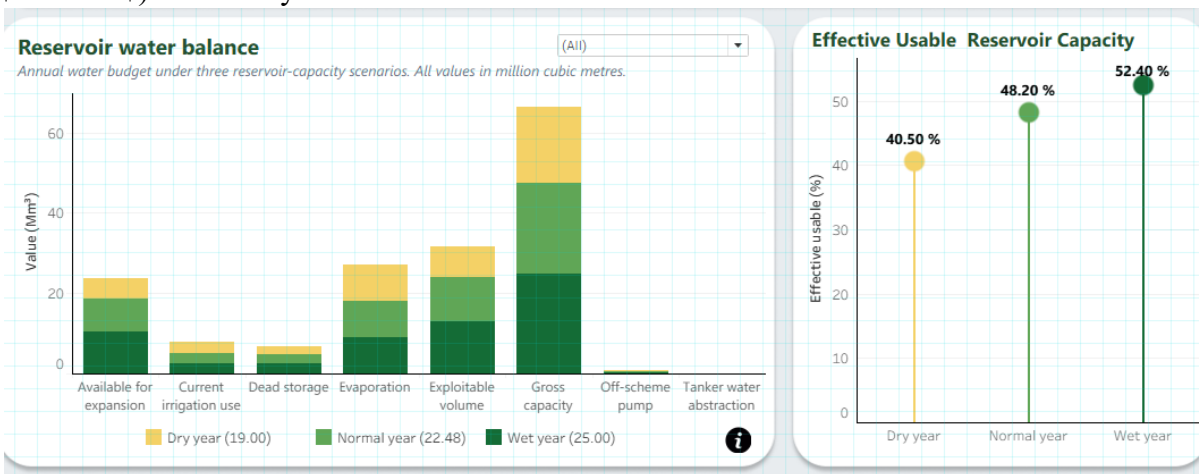
- **S1 (Rice only):** Expanding rice cultivation across both dry and wet seasons.
- **S2 (Rice and vegetables):** Splitting the new area between rice and dry-season vegetables.
- **S3 (Vegetables only):** Dedicating the new expansion entirely to a six-crop vegetable mix.
- **S4 (Improved efficiency):** Supporting rice production via wide-scale canal and infrastructure rehabilitation.
- **S5 (Off-scheme canal-pumped):** Growing specific high-value crops (pepper, okra, garden eggs) outside the main scheme boundaries using portable pumps.
- *User Action:* Each scenario includes a **"See More"** link to let users dive into deeper specific configurations.



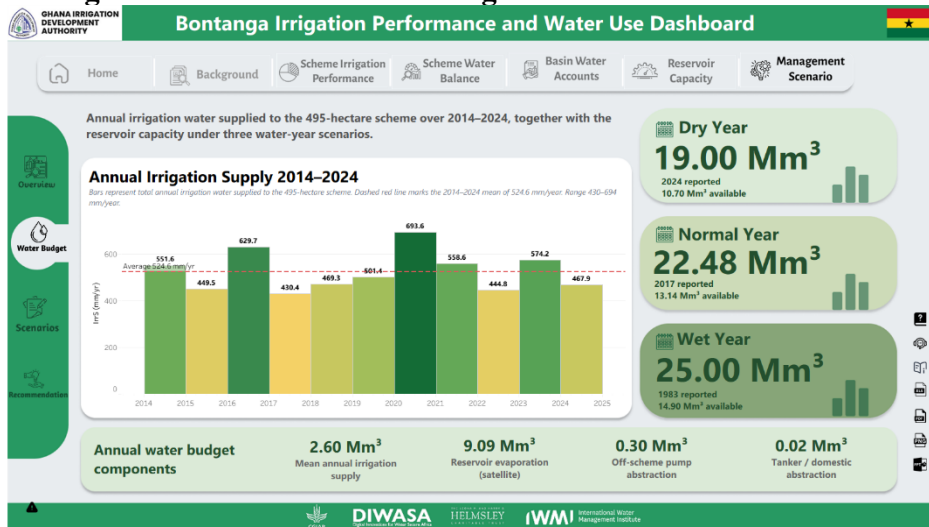
3. Climate Sensitivity Analysis (The Chart Panes)

Water management changes depending on weather patterns. The two bottom charts show users how the reservoir balances out under three distinct hydrological conditions: **Dry year** (yellow), **Normal year** (light green), and **Wet year** (dark green).

- **Reservoir Water Balance Chart:** A stacked bar chart that shows the user exactly how much water is consumed by evaporation, current irrigation, or dead storage, versus how much volume is left over and "Available for expansion."
- **Effective Usable Reservoir Capacity Graph:** A simple percentage plot showing users how much of the reservoir's total capacity remains practically usable during Dry (40.50%), Normal (48.20%), and Wet (52.40%) weather cycles.



Management Scenario: Water Budget Sub-Tab

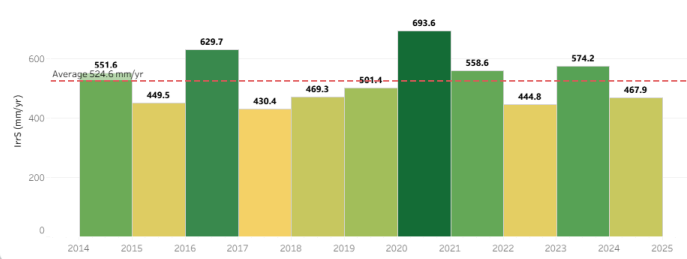


When a user selects the **Water Budget** option from the vertical menu on the left side of the dashboard, the page focuses entirely on evaluating historical water allocations alongside multi-scenario capacity benchmarks. It answers a fundamental user question: *“How much water do we historically supply, and what limits do different climate years impose on our reservoir capacity?”*

Annual irrigation water supplied to the 495-hectare scheme over 2014–2024, together with the reservoir capacity under three water-year scenarios.

Annual Irrigation Supply 2014–2024

Bars represent total annual irrigation water supplied to the 495-hectare scheme. Dashed red line marks the 2014–2024 mean of 524.6 mm/year. Range 430–694 mm/year.



historical average of **524.6 mm/yr**. Users can see instantly whether a particular year was above or below average.

What Can a User See and Interact With?

1. Annual Irrigation Supply Bar Chart (2014–2024)

This central chart shows users a ten-year historical trend of total annual irrigation water delivered to the 495-hectare scheme:

- **Trend Tracking:** Each vertical bar represents a single year's total supply (measured in mm/yr), color-coded to reflect supply volume drops or peaks (e.g., a peak supply of **693.6 mm/yr** in 2020 versus a dip to **430.4 mm/yr** in 2017).
- **The Baseline Mean:** A horizontal **dashed red line** cuts across the graph representing the 2014–2024

2. Three Water-Year Scenario Benchmarks (Right Sidebar)

On the right, users can review the absolute baseline capacity of the reservoir under three distinct historical climate conditions. This is essential for planning future expansions safely:

- **Dry Year Baseline:** Displays a total capacity of **19.00** (referencing 2024 data, with safely available).
- **Normal Year Baseline:** Displays a total capacity of **22.48** (referencing 2017 data, with safely available).
- **Wet Year Baseline:** Displays a total capacity of **25.00** (referencing 1983 data, with safely available).

3. Annual Water Budget Components (The Bottom Row)

This footer panel breaks down precisely where the basin's water volume is accounted for on average, helping users understand losses versus utilization:

- **Mean Annual Irrigation Supply: 2.60**
- **Reservoir Evaporation (Satellite Calculated): 9.09** (*Crucial for users to note, as evaporation constitutes a significant portion of natural water loss from the reservoir*).

- **Off-Scheme Pump Abstraction: 0.30**
- **Tanker / Domestic Abstraction: 0.02**

Key Summary for Users:

The Takeaway: Use this sub-tab to check historical supply consistency against environmental limits. By knowing that satellite-derived evaporation takes away roughly a year and that a Dry Year caps total capacity at , water planners can calculate exactly how much buffer room remains when designing new distribution schedules.

Annual water budget components

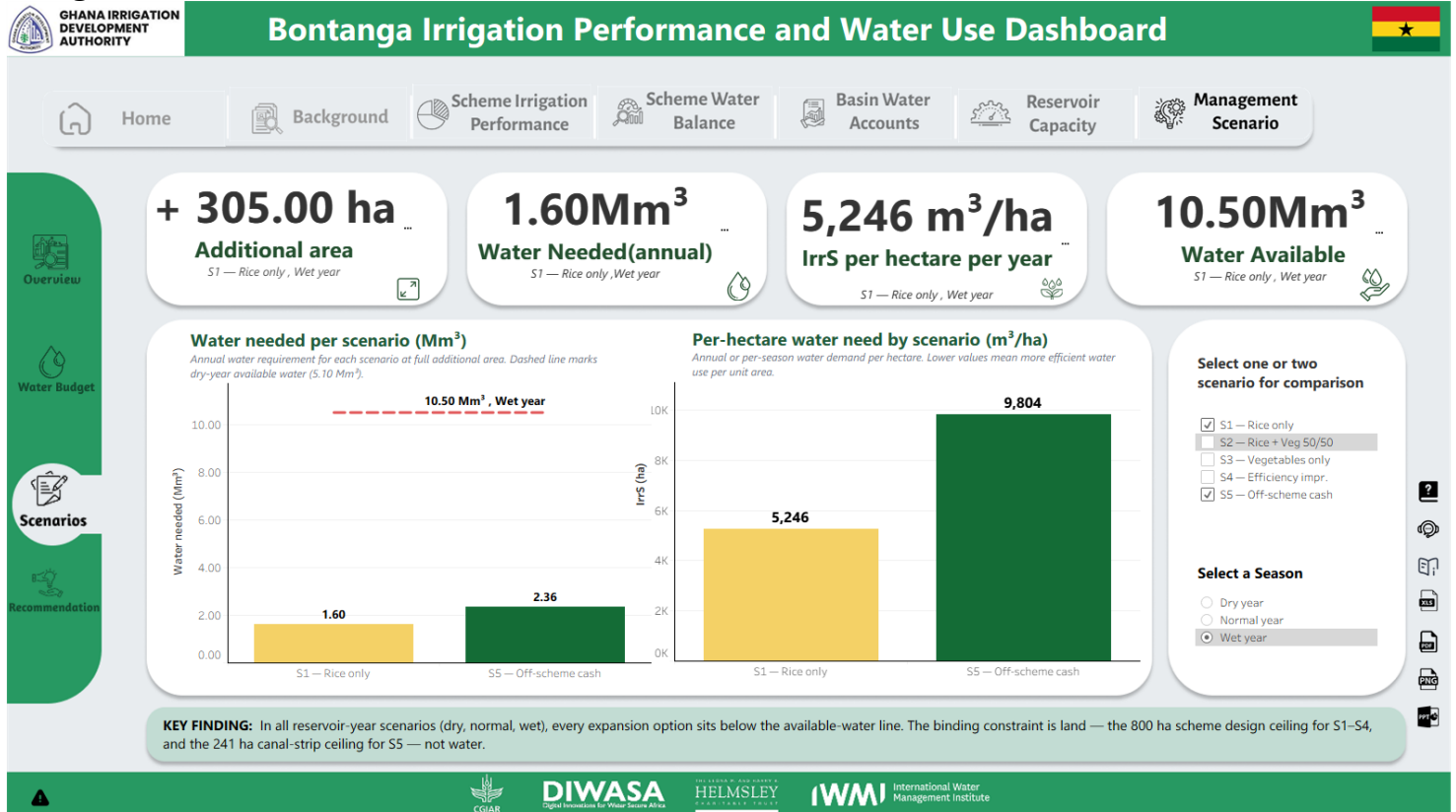
2.60 Mm³
Mean annual irrigation supply

9.09 Mm³
Reservoir evaporation (satellite)

0.30 Mm³
Off-scheme pump abstraction

0.02 Mm³
Tanker / domestic abstraction

Management Scenario: Scenarios Sub-Tab



When a user selects the **Scenarios** option from the vertical menu on the left side, the dashboard transforms into an interactive comparative modeling workspace. From a user's perspective, this page is the heart of the "What-If" analysis, allowing planners to pick and choose different expansion paths, compare them side-by-side, and immediately see the resource impacts.

Select one or two scenario for comparison

- ☒ S1 — Rice only
- ☐ S2 — Rice + Veg 50/50
- ☐ S3 — Vegetables only
- ☐ S4 — Efficiency impr.
- ☒ S5 — Off-scheme cash

Select a Season

- ☐ Dry year
- ☐ Normal year
- ☒ Wet year

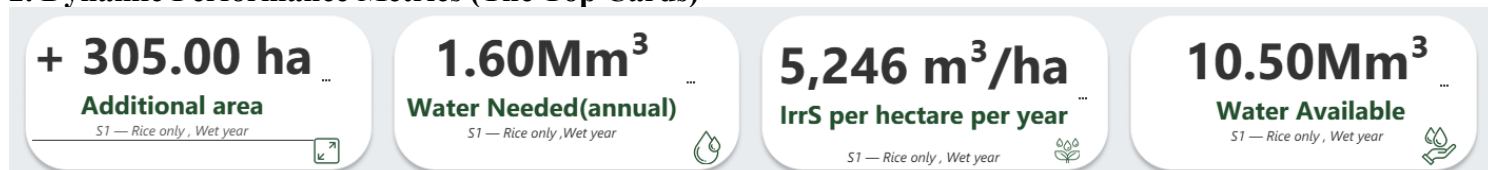
What Can a User See and Interact With?

1. Interactive Control Panel (The Right Sidebar)

This is where the user controls the entire page's simulation variables:

- Scenario Comparison Selection:** Users can check boxes to select **one or two scenarios** to contrast simultaneously (e.g., in image_16c2dd.jpg, S1 — Rice only and S5 — Off-scheme cash are selected).
- Season/Climate Filter:** A radio button selection allowing the user to toggle the simulation environment across three distinct weather cycles: **Dry year**, **Normal year**, or **Wet year** (with "Wet year" actively selected).

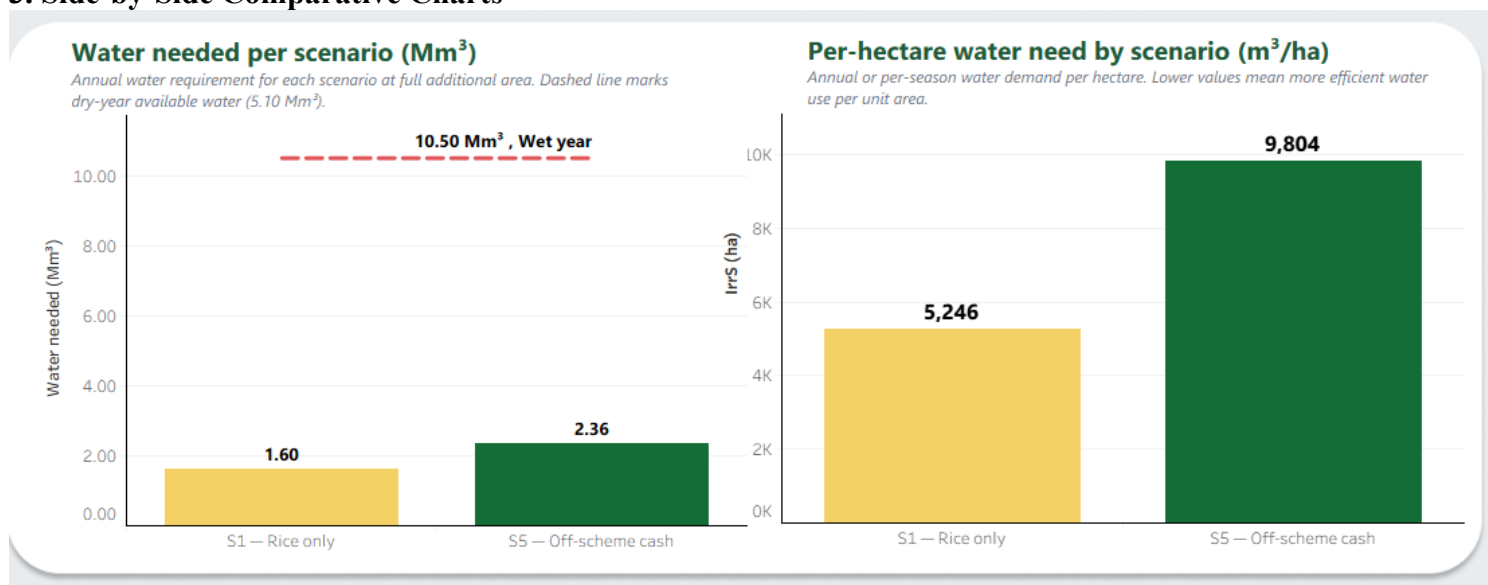
2. Dynamic Performance Metrics (The Top Cards)



These cards provide the immediate math results of the scenario settings selected on the right. They show small subtitles below the numbers (e.g., "S1 — Rice only, Wet year") to remind the user exactly what specific configuration the metric represents:

- **Additional Area:** Displays the total expansion footprint available (e.g., + **305.00 ha**).
- **Water Needed (Annual):** Shows the exact total volumetric demand required to sustain that scenario choice (e.g., **1.60 Mm^3**).
- **IrrS per Hectare per Year:** Breaks down the allocation intensity per individual hectare (e.g., **5,246 m^3/ha**).
- **Water Available:** Dynamically displays the absolute total water buffer available to the scheme under the chosen climate season (e.g., **10.50 Mm^3**).

3. Side-by-Side Comparative Charts



The central area visualizes the selected options against each other using intuitive bar charts:

- **Water Needed per Scenario (Mm^3):** This chart contrasts the gross annual water volume requirements. A vital user feature here is the **dashed red threshold line**, which explicitly marks the safe available water boundary. Users can visually confirm if a scenario bar stays safely below or breaches the limit.
- **Per-Hectare Water Need by Scenario (m^3/ha):** This chart isolates water efficiency. It lets users see which strategy is less water-demanding per unit area (e.g., showing that S1 uses a lower $5,246 \text{ m}^3/\text{ha}$ compared to S5's more intensive $9,804 \text{ m}^3/\text{ha}$).

4. The "Key Finding" Insights Box (Bottom Footer)

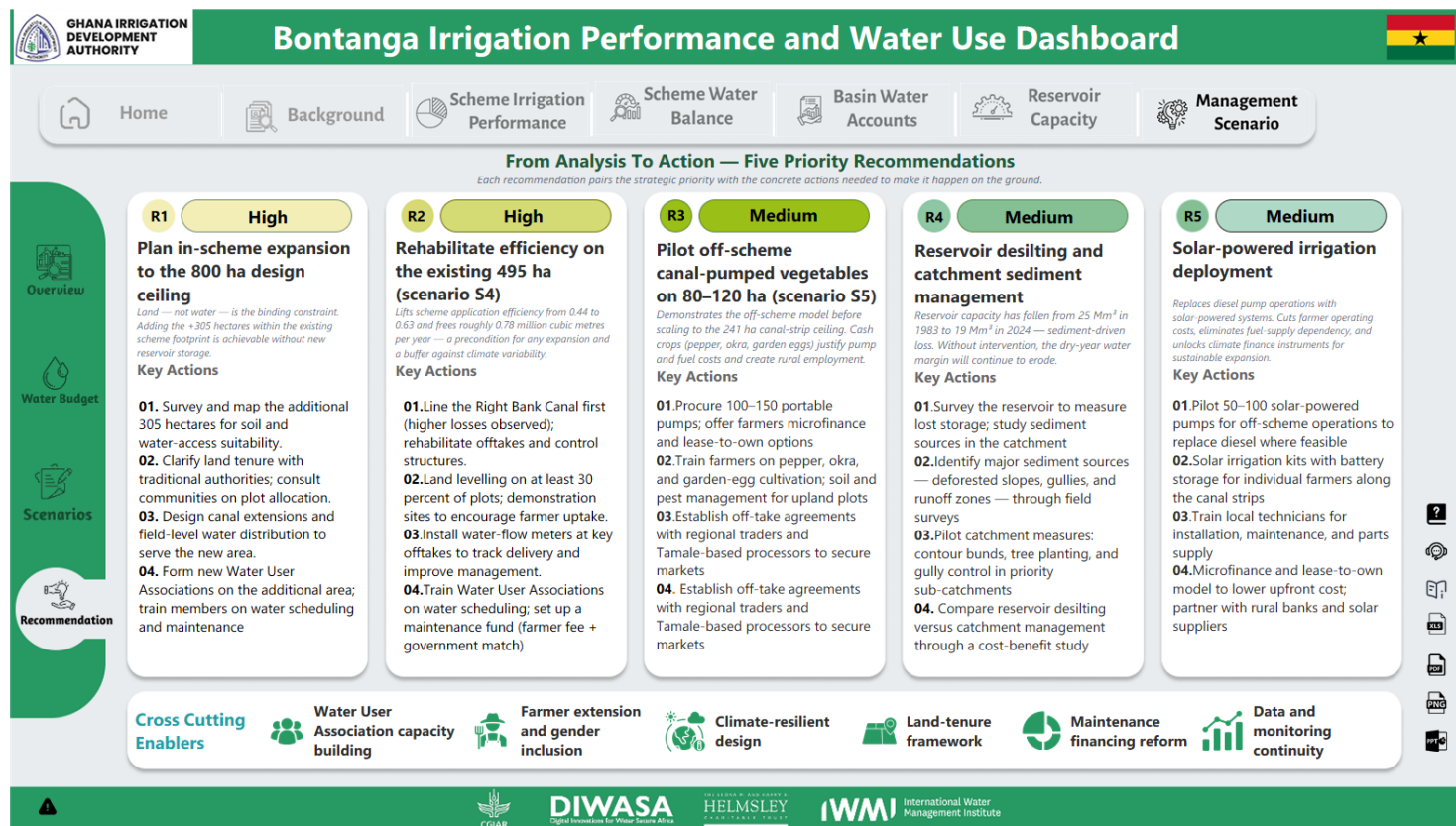
- A dedicated summary bar that highlights the core data takeaway for the user. It surfaces critical planning constraints—such as pointing out that across all weather variations, water availability is actually *not* the bottleneck; instead, physical land ceilings (**800 ha** for S1–S4 or **241 ha** for S5) form the absolute constraint.

KEY FINDING: In all reservoir-year scenarios (dry, normal, wet), every expansion option sits below the available-water line. The binding constraint is land — the 800 ha scheme design ceiling for S1–S4, and the 241 ha canal-strip ceiling for S5 — not water.

Key Summary for Users:

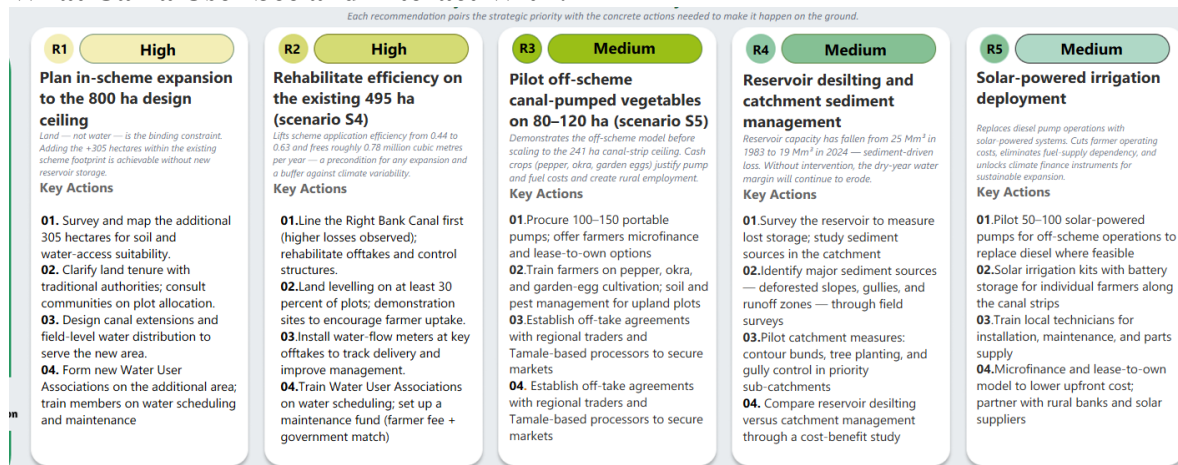
The Takeaway: Use this sub-tab to test combinations of agriculture expansions before making infrastructure choices. By checking any two options on the right, you can instantly see which choice uses water more efficiently per hectare, and ensure the gross volume required stays safely under the dashed red allocation line.

Management Scenario: Recommendation Sub-Tab



When a user selects the **Recommendation** option from the vertical menu on the left side, the dashboard shifts from modeling and comparison to concrete operational strategies. From a user's perspective, this page is the final action planning matrix, translating the data insights from previous screens into structured, prioritized project recommendations for decision-makers and engineers.

What Can a User See and Interact With?



1. Five Priority Recommendations (R1 to R5)

The main section of the page presents five detailed action cards. Each card is clearly tagged with an explicit **Priority Level (High or Medium)** and pairs a strategic irrigation objective with specific, step-by-step **Key Actions**:

- **R1: Plan in-scheme expansion to the 800 ha design ceiling (High Priority):** Focuses on clearing the land constraints by surveying the additional 305 hectares, clarifying land tenure with traditional authorities, designing canal extensions, and forming Water User Associations.
- **R2: Rehabilitate efficiency on the existing 495 ha (High Priority):** Focuses on lifting application efficiency from 0.44 to 0.63. Key actions include lining the Right Bank Canal, land levelling on at least 30% of plots, and installing water-flow meters.
- **R3: Pilot off-scheme canal-pumped vegetables on 80–120 ha (Medium Priority):** Focuses on executing the scenario S5 model for high-value cash crops. Actions include procuring 100–150 portable pumps, offering microfinance, and establishing off-take market agreements.
- **R4: Reservoir desilting and catchment sediment management (Medium Priority):** Addresses the historical loss of reservoir capacity due to sedimentation. Actions focus on surveying lost storage, piloting catchment controls (contour bunds, tree planting), and conducting a cost-benefit study on desilting.
- **R5: Solar-powered irrigation deployment (Medium Priority):** Promotes green energy by piloting 50–100 solar-powered pumps to replace diesel engines, offering lease-to-own financial models, and training local technicians.



2. Cross-Cutting Enablers (The Bottom Footer Panel)

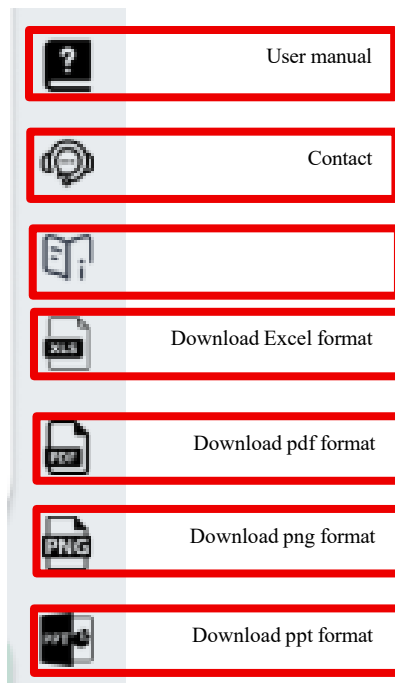
At the base of the page, users are presented with a series of institutional and programmatic pillars required to sustain the physical interventions above:

- **Water User Association capacity building**
- **Farmer extension and gender inclusion**
- **Climate-resilient design**
- **Land-tenure framework**
- **Maintenance financing reform**
- **Data and monitoring continuity**

Key Summary for Users:

The Takeaway: Use this sub-tab as an operational roadmap. Once you have used the **Scenarios** tab to understand the water impacts, come to this page to grab the ready-made action items, priority rankings, and institutional framework needed to pitch or execute the final expansion plan.

Additional features on the dashboard:



There are several additional features available on the dashboard. A brief description and purpose of each of the icons located on the lower left corner are presented here.

Contact: The contact icon is located on the lower right corner of the dashboard and it would provide email information on whom to contact in case you have any questions on the dashboard.

About: The about icon provides more info on the project.

Printing options: The dashboard can be printed or saved using three options. The current view of the dashboard can be saved to the local computer in three different formats – PDF, JPG or PPT and users can download data in excel format. Please use appropriate icon as per your need.